Eigendecomposition of Neural Networks
between NTK and Neural Collapse

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Abstract

While neural networks are incredibly effective at solving complex problems, the exact mathematical reasons behind how they learn remain an active area of research. To understand this, researchers often look at two leading theories. The first is the Neural Tangent Kernel (NTK), a framework that helps explain how networks evolve during training, and the second is Neural Collapse (NC), a phenomenon where networks perfectly separate and simplify data classes at the very end of the training process. However, the relationship between these two theories is still not entirely clear. In this study, we explore how these concepts interact by training models on a 2D canvas to classify distinct geometric shapes, such as circles, cardioids, stars, and astroids. We systematically test how different network sizes, depths, and initial setups affect the fundamental mathematical patterns (eigenfunctions) the model learns. Ultimately, we aim to connect the dots between how a network actively learns features and how it eventually clusters the data, revealing how a model's internal math responds to the geometric boundaries it is trying to learn.

1 Introduction

Modern deep learning theory proposes many approaches to explain the model training dynamics. Among the most popular ideas are infinite-width abstractions, that linearize neural network optimization trajectories, thus simplifying their analysis. Two foundational limits dominate the theory: the **Neural Network Gaussian Process (NNGP)** [7, 8], which characterizes the functional prior of infinitely wide networks at initialization, and the **Neural Tangent Kernel (NTK)** [6], which governs the deterministic evolution of networks under gradient descent and effectively linearizes the dynamics under the "lazy" training regime [2]. This is characterized by minimal weight and feature movement during training, complemented by a constant NTK. While these kernels offer strong theoretical guarantees, they stand in stark contrast to the **Feature Learning** regime (such as the Maximal Update Parameterization, or μP [14]), where representation learning actively changes the network's weights and the learned features during training, thus exiting the lazy regime and forcing the NTK to evolve dynamically during training.

Concurrently, deep networks trained past the point of zero training error (overparameterization) have been empirically shown to exhibit **Neural Collapse (NC)** at the terminal phase of training [10]. They tend to separate the data in classes, maximising the geometric distance in the feature space and minimizing the effect gradients of one class have on the features for other classes. This is mostly viewed through the lens of last-layer features contracting into a Simplex Equiangular Tight Frame (ETF) [10].

Recent work by Seleznova et al. [13] bridges the kernel interpretation and NC paradigms by demonstrating that Neural Collapse can be framed as a direct consequence of the structural evolution

of the NTK itself. Specifically, when the empirical NTK develops a block structure aligned with the class labels, the functional optimization dynamics under Mean Squared Error (MSE) loss lead to neural collapse. This result is based on the assumption that the NTK develops a block structure alligned with the classes, but then stays constent. While they work with classification tasks using mean squared error loss, this work adopts the more standard cross entropy loss and analyzes the continuous spatial dynamics of the NTK over a 2D domain and tests the fundamental constant block structure assumption. We also look at the NTK's eigenvalue spectrum and its corresponding eigenfunctions, which is crucial because they fundamentally govern the network's inductive bias and learning kinetics. Specifically, the eigenvalues dictate the exponential convergence rates of their respective eigenfunctions, directly illustrating how the network prioritizes learning low-frequency spatial patterns over high-frequency details (often referred to as spectral bias [11]).

By tracking spectral properties of the neural network, we systematically evaluate how different geometric target shapes and continuous decision boundaries warp the kernel, explicitly testing the block-structured kernel assumption during the progression toward a collapsed state. Furthermore, to isolate the pure geometric evolution of the network from Euclidean boundary artifacts (e.g., the Gibbs phenomenon [4,5]), we introduce a flat torus topological embedding ($S^1 \times S^1$). By contrasting standard bounded grids with this periodic, boundary-free space, we observe how initialization, parameterization, and topology dictate the spatial onset of neural collapse.

2 Definitions and Notation

2.1 Notation

Throughout this report, we adopt the following standard conventions: scalars are denoted by standard lowercase letters (e.g., x, λ), column vectors by bold lowercase letters (e.g., $\mathbf{x}, \mathbf{v}$), and matrices by bold uppercase letters (e.g., $\mathbf{X}, \mathbf{\Theta}$), while subscripts indicate elements of the vector (e.g., x_i). The trace and Frobenius norm of a matrix are denoted by $\text{Tr}(\mathbf{X})$ and $\|\mathbf{X}\|_F$, respectively. The Frobenius inner product between two matrices of the same dimension is defined as $\langle \mathbf{A}, \mathbf{B} \rangle_F = \text{Tr}(\mathbf{A}^T\mathbf{B})$.

The baseline input space is defined as a compact continuous 2D Euclidean domain $\mathcal{X} \subset \mathbb{R}^2$. To study topological boundary effects, we also define a transformation to a periodic flat torus embedding, denoted as $\mathcal{T} \subset \mathbb{R}^4$.

We consider an empirical dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ containing N samples, with input coordinates $\mathbf{x}_i \in \mathcal{X}$ and binary class labels $y_i \in \{0, 1\}$. For kernel alignment computations, it is often mathematically convenient to map these targets to a bipolar label vector $\mathbf{Y} \in \{-1, 1\}^N$, where $Y_i = 2y_i - 1$.

A neural network is expressed as a parameterized function $f(\mathbf{x}; \theta) : \mathbb{R}^d \to \mathbb{R}$, governed by parameters $\theta \in \mathbb{R}^P$. Analytical bivariate kernel functions mapping pairs of inputs to a scalar similarity are denoted as $K(\mathbf{x}, \mathbf{x}') : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$. Their corresponding empirical kernel matrices, evaluated over a discrete reference set of N points, are denoted by bold uppercase Greek or Roman letters (e.g., $\mathbf{\Theta} \in \mathbb{R}^{N \times N}$ specifically for the empirical NTK).

The eigendecomposition of an empirical kernel matrix yields a spectrum of non-negative eigenvalues sorted in descending order, $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_N \geq 0$, along with their corresponding orthonormal eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_N \in \mathbb{R}^N$. When an eigenvector is projected across the continuous spatial domain via the Nyström extension [9], the resulting continuous eigenfunction is denoted as $\psi_i(\mathbf{x})$.

2.2 Formal Definitions

Definition 2.1 (Neural Tangent Kernel & Block Structure [13]). *For a network output $f(\mathbf{x};\theta)$, the Neural Tangent Kernel $K_{NTK}(\mathbf{x},\mathbf{x}')$ is defined as the inner product of the network's functional gradients with respect to its parameters θ*

$$K_{NTK}(\mathbf{x},\mathbf{x}') = \left\langle \frac{\partial f(\mathbf{x};\theta)}{\partial \theta}, \frac{\partial f(\mathbf{x}';\theta)}{\partial \theta} \right\rangle.$$

Following [13], an empirical kernel matrix $\mathbf{\Theta} \in \mathbb{R}^{N\times N}$ displays an ideal label-aligned block structure if elements $\Theta_{ij} = K_{NTK}(\mathbf{x}_i,\mathbf{x}_j)$ satisfy

$$\Theta_{ij} = \begin{cases} \kappa_{in} & \text{if } y_i = y_j. \\ \kappa_{out} & \text{if } y_i \neq y_j. \end{cases}$$

where $\kappa_{in} > \kappa_{out} \geq 0$, creating a visible block structure in the label identity matrix.

While the NTK governs gradient descent dynamics, the Neural Network Gaussian Process (NNGP) characterizes the prior distribution of the network's outputs at initialization in the infinite-width limit.

Definition 2.2 (Neural Network Gaussian Process (NNGP) [7]). *For a network initialized with parameters drawn from a standard normal distribution, the NNGP kernel evaluates the expected covariance between two inputs as*

$$K_{NNGP}(\mathbf{x},\mathbf{x}') = \mathbb{E}_\theta \left[f(\mathbf{x};\theta) f(\mathbf{x}';\theta) \right].$$

Unlike the NTK, which actively evolves during feature learning, the NNGP represents the static functional prior of the chosen architecture.

Definition 2.3 (Kernel Alignment Score [3]). *To quantify the emergence of Neural Collapse and the formation of a label-aligned block structure, the Kernel Alignment Score A measures the cosine similarity between the empirical kernel matrix $\mathbf{\Theta}$ and the ideal target similarity matrix $\mathbf{Y}\mathbf{Y}^T$, using the Frobenius inner product*

$$A(\mathbf{\Theta},\mathbf{Y}) = \frac{\langle \mathbf{\Theta}, \mathbf{Y}\mathbf{Y}^T \rangle_F}{\|\mathbf{\Theta}\|_F \|\mathbf{Y}\mathbf{Y}^T\|_F}.$$

An alignment score approaching 1.0 indicates perfect separation into intra-class similarities and inter-class differences.

Definition 2.4 (Spectral Dominance and Effective Rank [12]). *The concentration of variance within the kernel spectrum is measured by Spectral Dominance (SD), defined as the ratio of the principal eigenvalue to the trace of the matrix*

$$SD(\mathbf{\Theta}) = \frac{\lambda_1}{Tr(\mathbf{\Theta})} = \frac{\lambda_1}{\sum_{i=1}^N \lambda_i}.$$

To measure the true dimensionality of the learned representation subspace, we treat the normalized eigenvalues $p_i = \frac{\lambda_i}{Tr(\mathbf{\Theta})}$ as a probability distribution and define the Effective Rank (ER) as the exponentiated Shannon entropy

$$ER(\mathbf{\Theta}) = \exp\left(-\sum_{i=1}^N p_i \log p_i \right).$$

A collapsing network will exhibit increasing spectral dominance and a rapidly decreasing effective rank, theoretically lower-bounded by the number of distinct classes.

Definition 2.5 (Kernel Eigenfunctions via Nyström Extension)**.** *Given an empirical kernel matrix* $\mathbf{K} \in \mathbb{R}^{N \times N}$ *evaluated on reference points* $\mathbf{X}_{eval}$ *with eigenvalues* λ_i *and eigenvectors* $\mathbf{v}_i$*, the continuous eigenfunction* $\psi_i(\mathbf{x})$ *evaluated at an arbitrary grid coordinate* $\mathbf{x} \in \mathcal{X}$ *is approximated via the Nyström projection*

$$\psi_i(\mathbf{x}) = \frac{1}{\lambda_i} \sum_{j=1}^{N_{eval}} K(\mathbf{x}, \mathbf{x}_j) v_{ij}.$$

3 Methodology: The Classification Problem

To systematically study how neural networks resolve continuous spatial boundaries during Neural Collapse, we must separate the geometry of the target classification task from the topology of the input space.

3.1 Continuous Decision Boundaries

We frame the problem as the task of learning explicit geometric shapes within a continuous 2D plane. We define continuous margin functions for a variety of shapes, spanning simple convex shapes to complex, non-linear boundaries, and even the presence of two separate shapes in the same domain. The primary shapes evaluated include a circle, heart, star, astroid, and a disjoint "XOR circles" configuration. Figure 1 provides a visual overview of the exact boundaries for each of these target geometries. To satisfy the balancing requirements necessary to study Neural Collapse cleanly, boundaries are explicitly derived to partition the domain into equivalent areas. This allows us to sample points from the domain with almost perfect class balance. Table 1 shows the shapes and formulas we used.

Shape	**Formula**	**Properties**
Circle	$y = \mathbb{I}\left(x_1^2 + x_2^2 > 2.546\right)$	A smooth baseline mapping.
Heart (Cardioid)	$y = \mathbb{I}\left((0.77x_1^2 + 0.77x_2^2 - 1)^3 - 0.77^2 x_1^2 \cdot 0.77^3 x_2^3 < 0\right)$	An asymmetric, single sharp indentation, smooth lobes.
Star (5-Petal)	$y = \mathbb{I}\left(\sqrt{x_1^2 + x_2^2} > 1.54 + 0.4\cos(5 \cdot \operatorname{atan2}(x_2, x_1))\right)$	A periodic shape.
Astroid	$y = \mathbb{I}\left(\lvert x_1\rvert^{2/3} + \lvert x_2\rvert^{2/3} > 2.603^{2/3}\right)$	A hypocycloid, four highly nonlinear, orthogonal sharp cusps.
XOR Circles	$y = \mathbb{I}\left(\min\left((x_1 - 2.0)^2 + (x_2 - 2.0)^2, (x_1 + 2.0)^2 + (x_2 + 2.0)^2\right) > 5.0\right)$	A disconnected shape for non torus case, circle for torus

Table 1: Summary of target domain topologies, their corresponding level-set indicator formulas, and structural properties.

By defining these shapes via exact analytical formulas, we establish a rigorous ground truth for continuous eigenfunction evaluation, allowing us to map the functional behavior of the Neural Tangent Kernel across every coordinate in the domain and to qualitatively evaluate how the eigenfunctions of the NTK fit these shapes.

3.2 Feature Engineering

Neural networks trained on bounded Euclidean domains frequently suffer from boundary artifacts. As we show in the evaluations of the NTK spectrum, forcing an eigenfunction to truncate at the

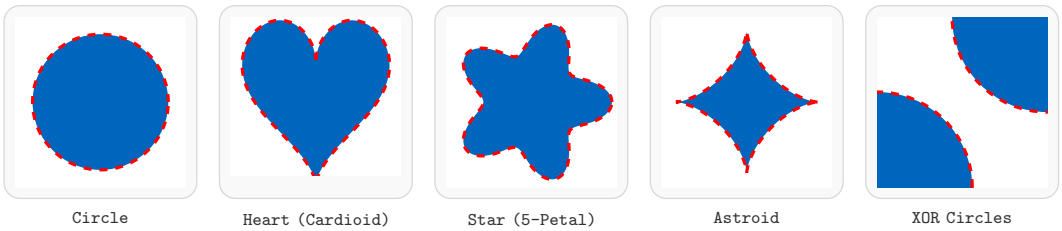

Figure 1: Visual overview of the continuous 2D target geometries evaluated in this study. The red dashed lines represent the theoretical boundaries separating the balanced binary classes.

domain limits induces high-frequency ringing in the top eigenfunctions, analogous to the Gibbs phenomenon, thus complicating the analysis. This is why we evaluate two distinct input topologies to isolate these effects and make our findings more robust to this phenomenon.

1. Standard Euclidean Grid ($\mathbb{R}^2$)

In the baseline, the input space is defined as the bounded square region $\mathcal{X} = [-2, 2] \times [-2, 2]$. Inputs $\mathbf{x} = (x_1, x_2) \in \mathcal{X}$ are fed directly into the first layer of the network. Because the domain is bounded and the standard Euclidean distance metric is used, the network is forced to learn the artificial boundaries of the grid alongside the true geometry of the target shape.

2. Flat Torus Projection ($S^1 \times S^1 \subset \mathbb{R}^4$)

To eliminate boundary-induced artifacts and observe the pure geometric evolution of the network, we introduce a periodic topological embedding. We map the bounded 2D square onto a flat torus manifold. To achieve this, the input coordinates are scaled to the interval $[-\pi, \pi]$ and projected into $\mathbb{R}^4$ using trigonometric functions:

$$\Phi_{\text{torus}}(x_1, x_2) = \left(\cos\left(\frac{\pi}{2}x_1\right), \sin\left(\frac{\pi}{2}x_1\right), \cos\left(\frac{\pi}{2}x_2\right), \sin\left(\frac{\pi}{2}x_2\right)\right).$$

This mapping is bijective over the interior of $\mathcal{X}$ and explicitly identifies opposite edges of the domain with one another. Unlike a standard 3D torus embedding which distorts distances due to inherent curvature, this 4D embedding creates a *flat* torus, perfectly preserving the local flat geometry of the space while seamlessly wrapping the domain into a periodic, boundary-free manifold.

To ensure a rigorous comparison between the standard network and the torus-embedded network, we computationally separate the target label generation from the feature projection. The ground-truth continuous margins and resulting binary class labels y_i are computed entirely in the standard 2D Euclidean space for all experiments. The projection Φ_{torus} is treated purely as a feature engineering step applied immediately prior to the network input. Consequently, the $\mathbb{R}^4$ torus network is forced to approximate the exact same 2D Euclidean shape target as the standard $\mathbb{R}^2$ network. This deliberate misalignment between the periodic inductive bias of the torus network and the non-periodic Euclidean targets enables us to directly measure the impact of boundaries on the spatial dynamics of Neural Collapse and the eigenfunctions of the NTK.

4 Experimental Setup

To evaluate the interplay between network parameterization, initialization, and the decision boundary during the happening of Neural Collapse, we establish a controlled experimental framework across all target geometries.

4.1 Architectures and Parameterizations

We examine three distinct parameterization regimes to contrast lazy training dynamics against active feature learning:

- **Standard Parameterization:** Conventional initialization where hidden weights are typically drawn from a normal distribution and scaled by a factor of $1/\sqrt{d}$ (for the input layer of dimension d) and $1/\sqrt{n}$ (for hidden layers of width n). The learning rate remains $\mathcal{O}(1)$ uniformly across all layers. While this maintains stable variance across hidden representations at initialization, the weight updates and feature learning inherently vanish at a rate of $1/\sqrt{n}$ as the network width approaches infinity.
- **Neural Tangent Kernel (NTK) Parameterization:** Initialized and scaled to explicitly operate in the infinite-width lazy training regime. Weights are typically drawn from a standard normal distribution $\mathcal{N}(0, 1)$ and are explicitly multiplied by $1/\sqrt{n}$ during the forward pass. Combined with a uniform $\mathcal{O}(1)$ learning rate, this exact scaling restricts weight updates to an infinitesimal $\mathcal{O}(1/\sqrt{n})$ distance from their initial values. Consequently, internal representations remain nearly static and the model acts as a fixed linear kernel predictor governed solely by its initial NTK [6].
- **Maximal Update Parameterization (μP):** Structured such that hidden representations evolve non-trivially ($\mathcal{O}(1)$ updates) even in the infinite-width limit, allowing the network to actively perform feature learning. To achieve this, μP introduces decoupled, layer-specific scaling: hidden weights are scaled by $1/\sqrt{n}$ at initialization, but the output layer weights are aggressively scaled down by $1/n$. To compensate and drive active learning, the learning rates for the hidden and output layers are scaled up by a factor of $\mathcal{O}(n)$, while the input layer learning rate remains $\mathcal{O}(1)$. This ensures the network learns abstract features rather than remaining purely kernel-based [14]. We will reference this as max parametrization in the rest of this report.

All networks are constructed with configurable hidden widths $n \in \{10, 100, 1000\}$ and depth configurations of $L \in \{1, 2\}$ hidden layers, utilizing the Rectified Linear Unit (ReLU) activation function. Models are trained using stochastic gradient descent, optimizing over binary cross entropy loss. We use Stochastic Gradient Descent (SGD) with a fixed learning rate $\eta = 1.0$ for 1000 epochs.

4.2 Evaluation and Logging

Evaluation is conducted periodically throughout training to track temporal and spatial evolution of the system. We record:

- **Kernel Values**: Kernel heatmap evaluated periodically through training on train and validation sets, with points sorted according to distance to the theoretical decision boundary to observe the block structure.
- **Kernel Spectra and Alignment:** Empirical NTK and NNGP matrices tracking the Kernel Alignment Score (A), Spectral Dominance (SD), and Effective Rank (ER).
- **Continuous Eigenfunctions:** Nyström extensions evaluated across a dense 200×200 Cartesian grid ($\mathcal{X}$) and a 1D diagonal slice to visualize spatial harmonic evolution without boundary distortion and a clear ordering of the points in the continuous domain.

All tracking, metric logging, and artifact generation are managed through Weights & Biases [1] (wandb) to ensure reproducible tracking across parameterization and shape sweeps.

5 Results

In this section, we present the empirical results of our analysis across different domain topologies and parametrizations. Unless explicitly stated, all experiments and figures utilize the standard parametrization and a network architecture consisting of two hidden layers with 1000 neurons.

5.1 Periodic vs. Bounded Domain Constraints

Here, we want to motivate the torus domain change by highlighting the spectral differences of similar models trained on the same task in the periodic and bounded domains. We will look at this through the eigenvalue spectra of the networks and how they decay, and the top four eigenfunctions of the NTK. For this section, all of the figures belong to networks that use the standard parametrization.

5.1.1 Boundary Differences

Figure 2 (left side) shows how the choice of boundary affects the first NTK eigenfunction (Ψ_0). With standard bounded constraints (No Torus), strange artifacts appear. Specifically, the eigenfunction is pushed heavily into the top-right corner instead of centering on the data. This shows that hard boundaries force the model to react to the edges of the space, distorting how it sees the Astroid. In contrast, using a periodic domain (Torus) fixes this issue. The Torus eigenfunction a full bump inside the Astroid, while this is not visible in the bounded case. This demonstrates that removing hard boundaries allows the network to learn the true shape without edge interference.

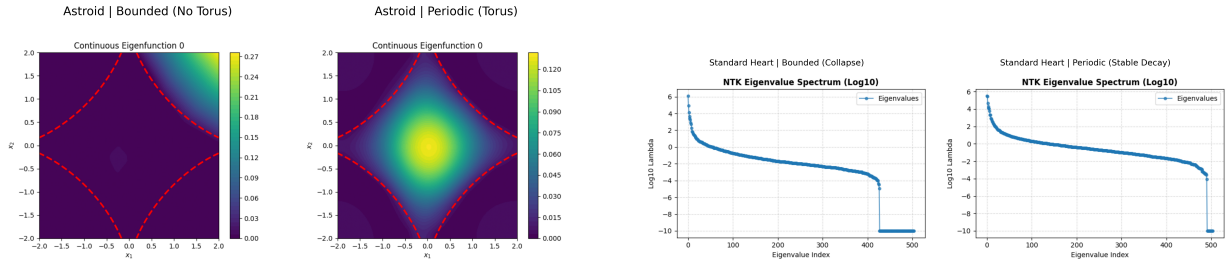

Figure 2: Effects of domain constraints at $W = 1000$. **Left:** Comparison of the fundamental eigenfunction (λ_0) for the Astroid geometry under bounded versus periodic (torus) domains. **Right:** Eigenvalue spectrum for the Heart shape under standard parametrization for bounded (square) and periodic (torus) models.

Figure 2 (right side) illustrates the impact of boundary conditions on the network's spectral properties by comparing the Neural Tangent Kernel (NTK) eigenvalue spectra across bounded and periodic domains. In the bounded configuration, the spectrum exhibits a steady decay before undergoing an abrupt collapse just after the 400th eigenvalue index, plummeting to a numerical floor of $\log_{10} \lambda = -10$. This could be the effect of the hard boundaries that restrict the network's effective capacity, abruptly cutting off its ability to utilize higher-order features. Conversely, the periodic Torus domain extends this regime of stable decay, postponing the severe spectral collapse until approximately the 490th index. By sustaining a broader sequence of active eigenvalues, the periodic topology enables the network to leverage a richer, higher-dimensional feature space to represent the target shape before reaching absolute numerical limits.
Figure 3 highlights how the boundary choice determines whether the XOR Circles shape acts as a connected or disconnected geometry. In the standard bounded space (No Torus), the data regions

in opposite corners are physically separated from each other. Because of this disconnection, the first four eigenfunctions simply push outward into the isolated corners, treating the data as fragmented and completely separate clusters. When the periodic Torus domain is applied, the edges of the space wrap around and connect. This links the previously separated corner regions into a single continuous shape. As a result, the Torus eigenfunctions form unified, oscillating patterns that flow smoothly across the newly connected clusters. This demonstrates that the periodic boundary fundamentally changes the problem, allowing the model to learn a unified geometric structure rather than isolated pieces.

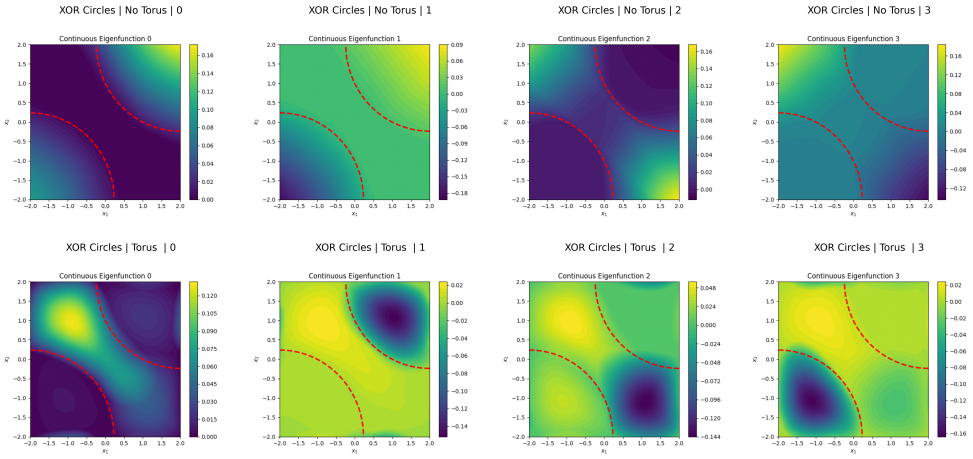

Figure 3: Comparison of the first four eigenfunctions ($\lambda_0 \dots \lambda_3$) for the XOR Circles distribution under bounded versus periodic (torus) domain constraints at $W = 1000$.

5.1.2 Geometric Complexity: Circle vs XOR

Figure 4 demonstrates how domain topology alters the network's perception of geometric complexity. In the standard bounded space, the contiguous Circle and the disjoint XOR shapes produce clearly different eigenvalue spectra, meaning the network is forced to treat them as two entirely separate and distinct learning problems. However, under the periodic Torus constraints, the spectra for both the Circle and the XOR shapes become nearly identical. This occurs because, on a wrapping Torus domain, the disjoint XOR regions connect across the boundaries. As a result, the Torus boundary removes the artificial complexity of the disjoint layout, allowing the network to process both shapes using a similar spectral structure.

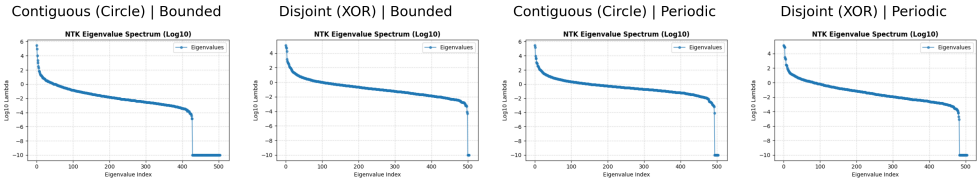

Figure 4: Comparison of the standard eigenvalue spectrum for the contiguous Circle versus the disjoint XOR shapes under periodic (torus) domain constraints at $W = 1000$. This shows how the XOR shape connects through boundaries of the torus manifold to transform the problem to a translated circle shape.

5.2 NTK vs Max Parametrization

Now that we motivated the choice of the periodic domain, we are ready to explore the differences between the lazy kernel regime (NTK parametrization) and the feature learning regime (max parametrization). This will be shown with the help of effective rank, difference from initialization, and NTK heatmaps.

5.2.1 Effective Rank and Difference from Initialization

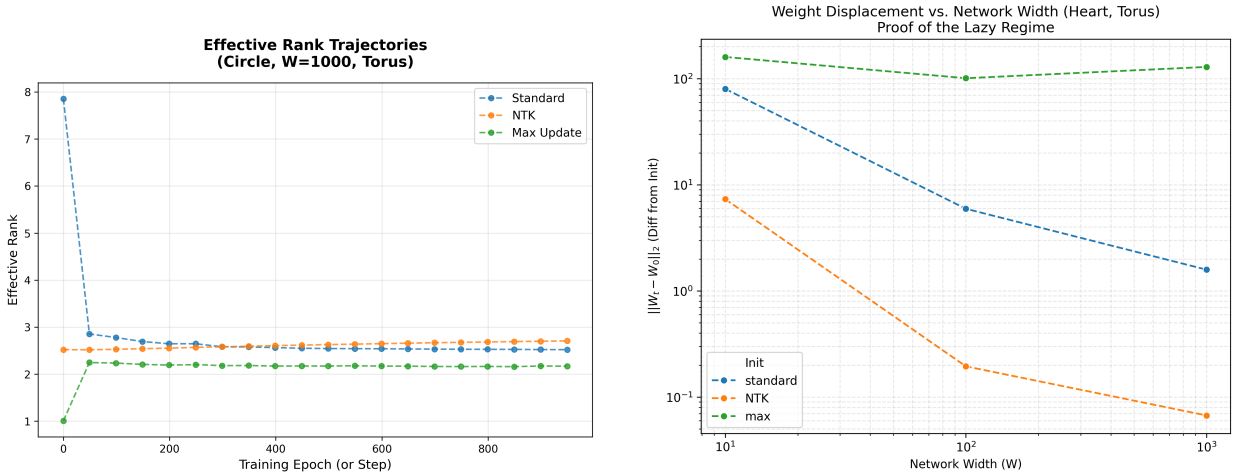

Figure 5: Spectral and displacement dynamics in wide networks (Torus). **Left:** Trajectories of the effective rank over training epochs for the contiguous Circle geometry at $W = 1000$. The Standard parameterization exhibits distinct spectral collapse, Max maintains extreme compression, and NTK remains fixed in a higher-rank state. **Right:** difference from initialization ($||K_t - K_0||_2$) as a function of network width (W) for the Heart geometry. NTK parameterization converges to near-zero displacement (lazy regime) at extreme widths, whereas the Max parameterization maintains active feature learning, with the standard parametrization being in between.

Figure 5 highlights clear differences in how various parametrizations handle effective rank, both at initialization and over the course of training. At the start of training, the models begin at entirely different structural states. The Standard parametrization starts with a very high effective rank (near 8) because standard random initialization creates a wide spread of scattered, uncorrelated features. The NTK starts at a moderate rank (around 2.5) because its specific scaling stabilizes the initial variance, and it remains almost entirely flat throughout training, showing that its representation barely moves and avoids compressing the spectrum. Most notably, the Max Update parametrization starts with an effective rank of exactly 1. This extremely low starting point occurs because its scaling deliberately suppresses the initial random variance, forcing the network into a simple state so it must build its features entirely from scratch. As training progresses, the Standard parametrization experiences a sharp initial drop from its high starting point but ultimately settles at a higher effective rank than the Max Update model. The Max Update parametrization maintains a tightly compressed spectrum and finishes with the lowest overall effective rank. This confirms that the Max Update regime engages in active feature learning, condensing the data representation down to its most essential components.

Figure 5 illustrates the NTK's difference from initialization ($||K_t - K_0||_2$) as network width increases. As the width grows to $W = 1000$, the difference from initialization for the NTK

parametrization drops dramatically to near zero (below 10^{-1}), indicating that it operates purely in the "lazy regime," meaning its weights barely move from their initial state during training. In contrast, the max parametrization maintains a consistently high difference from initialization (remaining above 10^2) across all widths, proving that it continues to actively learn and adjust its features. The standard parametrization exhibits intermediate behavior; while its displacement decreases as width increases, it remains significantly higher than the NTK model at $W = 1000$, visually placing it in a transitional space between the frozen lazy regime and the rich feature-learning regime of the max update.

5.2.2 NTK Block Structure

Figure 6 demonstrates the evolution of the Frobenius-normalized NTK matrices, ordered by distance to the decision boundary. For the `standard` and `max` initializations, the kernel actively transitions from initialization noise to a stark block-diagonal structure by the end of training. The bright diagonal blocks indicate intra-class similarity collapse, where points deep within the same class region become highly correlated, while the dark off-diagonal areas show active suppression of cross-boundary similarity. This behavior provides direct empirical validation of the block-diagonal structure required by the tangent neural collapse framework [13]. In contrast, the `NTK` matrix remains frozen in its unorganized, noisy initial state, confirming that the lazy regime prevents the network from dynamically aligning its internal representations with the target geometry. Furthermore, the kernel similarity peaks distinctly in the intra-class corners, representing points furthest from the decision boundary and decays drastically as the coordinates approach the boundary lines. This spatial gradient within the blocks demonstrates that convergence behavior varies across subsets of the continuous domain, with the network maximizing feature collapse deep within class regions while actively preserving representational variance near the classification boundary.

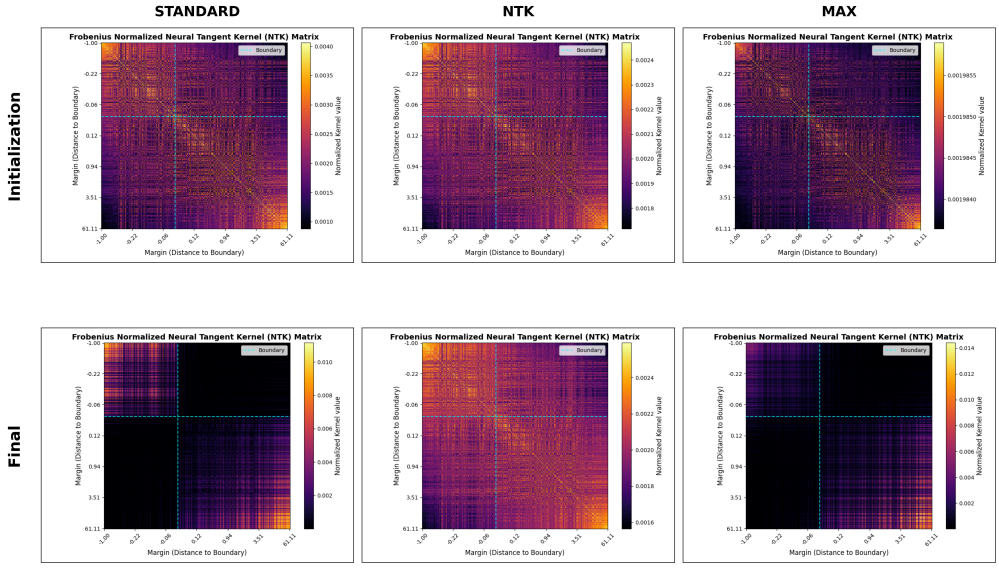

Figure 6: Evolution of the NTK matrix ordered by boundary distance. The `standard` and `max` initializations actively develop a high-contrast block structure, whereas the `NTK` initialization remains statically noisy. This also motivates the continuous domain choice, as the areas farther from decision boundaries are purer similarity-wise.

5.3 Eigenfunction Roles and Differences

In this part, we focus on the NTK eigenfunctions. We analyze the differences between the first four eigenfunctions and try to group them and uncover the link to the underlying decision boundary they try to approximate across different parametrizations and shapes. We will also look at diagonal eigenfunctions, which are the eigenfunctions of an NTK solely computed on points of the first diagonal of the domain, formally $\{x, y \in [-2, 2], \quad x = y\}$. These are one dimensional functions and are therefore easier to analyze. Since we work on the torus domain, we expect all eigenfunctions to be periodic functions.

5.3.1 Lazy Regime vs Feature Learning

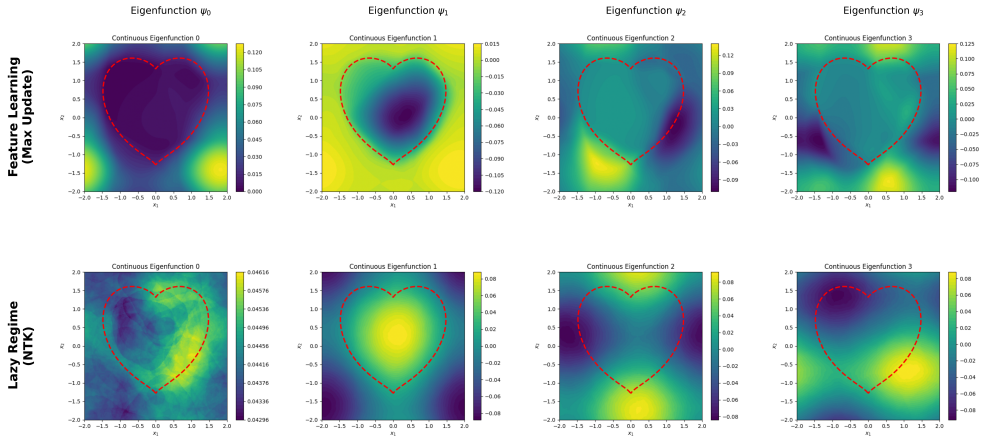

Figure 7: Comparison of the fundamental eigenfunction ($\Psi_{0...3}$) for the Heart geometriy under NTK and maximal update parametrizations at $W = 1000$.

Figure 7 contrasts the spatial resolution of the first four continuous eigenfunctions between the feature-learning and lazy regimes. Under the `max` update parametrization (top row), the network actively sculpts its eigenfunctions to align with the target geometry. The primary eigenfunction (ψ_0) develops a clear, high-contrast separation between the interior and exterior of the Heart shape, while the higher-order modes (ψ_1 through ψ_3) partition the space into distinct, purposeful sub-regions with well-defined, smooth gradients. This visualizes a rich feature-learning regime where weights dynamically adjust to cleanly resolve the decision boundary.
In contrast, the `NTK` parametrization (bottom row) exhibits the visual hallmarks of the lazy regime. The principal eigenfunction (ψ_0) is severely degraded by unstructured, high-frequency initialization noise, entirely failing to distinctly map the target shape. Although the subsequent modes (ψ_1 through ψ_3) display generic, low-frequency spatial patterns, they lack the sharp, target-aware structural alignment seen in the active network. Because the weights remain effectively frozen, the `NTK` model cannot actively refine these random initial projections to fit the specific spatial geometry of the boundary, forcing it to rely on generic basis functions corrupted by initial noise artifacts.

5.3.2 Diagonal eigenfunctions

Figure 8 illustrates the evolution of the top four diagonal eigenfunctions along the diagonal of the domain for the Circle target. In the bounded domain, the `standard` initialization produces rigid, piecewise linear splines that abruptly flatline or kink precisely at the spatial boundary markers. This

truncation reflects a saturated spatial confidence, where the network exhausts its representational capacity modeling hard edges, mechanically triggering the observed spectral collapse. Conversely, the periodic Torus domain removes these artificial spatial constraints, allowing the `standard` and `max` networks to develop smooth, continuous sinusoidal waves that transition gradually across the boundary. This translation-invariant, Fourier-like feature extraction preserves the network's expressive capacity and stabilizes spectral decay. Furthermore, contrasting the `standard` and `NTK` parameterizations in the periodic domain highlights the limitations of the lazy regime; while `max` networks actively smooth their representations to map the boundary, the `NTK` eigenfunctions remain dominated by high-frequency initialization noise and almost constant eigenfunctions, demonstrating an inability to actively organize their spatial geometry.

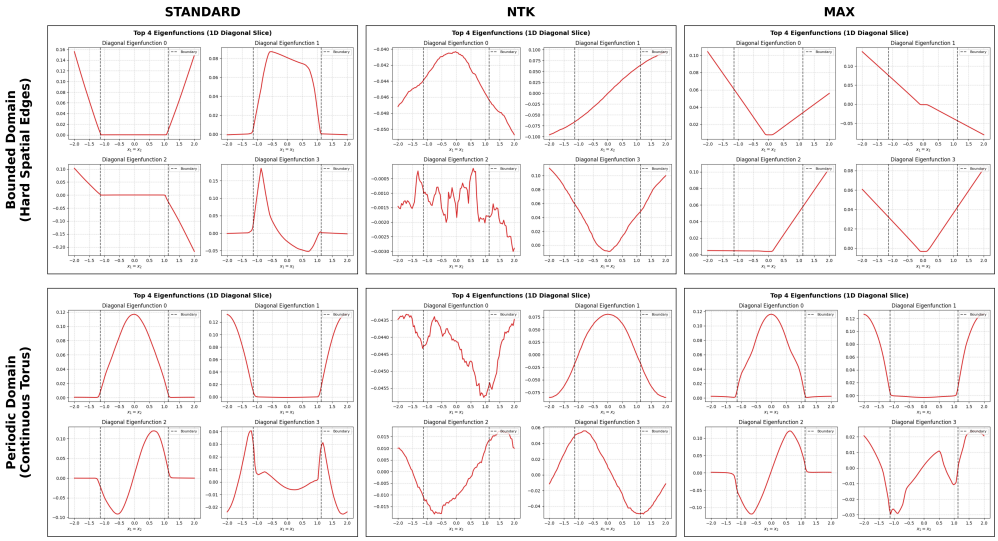

Figure 8: Evolution of the top four diagonal eigenfunctions with respect to the decision boundary distance for the Circle target. All evaluated samples are on the first diagonal of the domain.

5.4 Neural Collapse

Now we can try to establish a link between the different parametrizations and neural collapse. We do that by exploring NC1 and NC2 metrics across identical models differently parametrized. Figure 9 examines the neural collapse phenomenon by tracking intra-class and inter-class variance during training for the Circle geometry. The NTK regime remains structurally static, showing completely flat trajectories for both metrics, which confirms that its internal representations remain frozen and it does not actively cluster the data. In contrast, the Max Update and Standard parameterizations demonstrate active, albeit distinct, representational clustering and class separation dynamics. The Max Update parametrization maintains extremely low intra-class variance while drastically increasing its inter-class variance, indicating that it actively pushes different classes apart while tightly compressing the representations within the same class.

This distinction between the lazy and active regimes is further corroborated by the Kernel-Target Alignment (KTA) shown in Figure 9. Over the training epochs for the XOR Circles distribution, the NTK remains perfectly static, showing no improvement in how well its kernel aligns with the target labels. Meanwhile, the feature learning regimes—particularly the Max Update model—experience a massive surge in alignment, proving that active feature adaptation is essential for achieving structural collapse and optimal class separation.

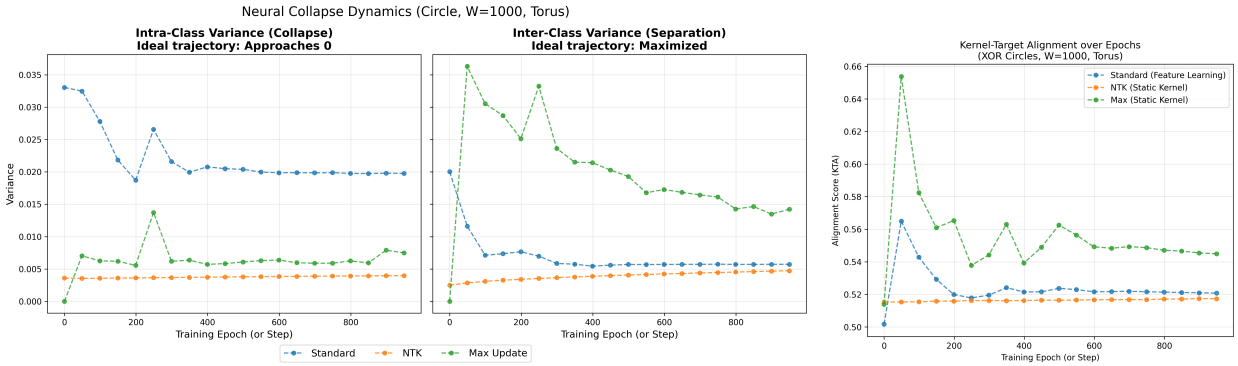

Figure 9: Feature adaptation and clustering dynamics during training at $W = 1000$ (Torus). **Left:** Evolution of intra-class and inter-class variance for the Circle geometry. The NTK regime remains structurally static, while the Max and Standard parameterizations demonstrate active, albeit distinct, representational clustering and class separation dynamics. **Right:** Evolution of Kernel-Target Alignment (KTA) over training epochs for the XOR Circles distribution, demonstrating the static nature of the NTK versus active feature adaptation in the Max and Standard parameterizations.

6 Conclusion

This study explores how the Neural Tangent Kernel (NTK) and Neural Collapse (NC) interact within continuous 2D spaces under varying boundary conditions and network parameterizations. By utilizing a flat torus, we demonstrated that standard bounded Euclidean grids artificially distort what a neural network learns. Hard boundaries force the network's eigenfunctions into the corners of the space and artificially fragment inherently connected, yet spatially disjoint, shapes like the XOR circles. Conversely, the boundary-free torus manifold allows the network to capture the true underlying geometry of the data, processing both connected and disconnected geometries with an identical spectral structure.

Furthermore, our comparison between parameterizations contrasts lazy training with active feature learning. In wide networks, the NTK regime remains entirely static: its weights barely move, its effective rank avoids compression, and its eigenfunctions ignore the actual geometric boundaries of the data. In contrast, the Maximal Update Parameterization actively engages in feature learning. Its weights adapt significantly, it tightly compresses the eigenvalue spectrum, and it actively molds its internal representations to fit the data boundaries.

Additionally, these distinct spatial and spectral behaviors dictate whether Neural Collapse can actually occur. Because the NTK regime is static, it fails to cluster the data; its intra-class and inter-class variances remain flat, and its alignment with target labels shows no improvement. True Neural Collapse, where data points of the same class group tightly together while different classes are pushed far apart, only emerges during active feature learning. This confirms that Neural Collapse is not merely a byproduct of late-stage optimization, but a direct consequence of the network actively adapting to the geometry of the data.

While this work establishes a foundational understanding of NC and the NTK in periodic and bounded 2D domains, several avenues remain for future investigation. A primary direction is to investigate how neural collapse unfolds spatially across the continuous domain, particularly concerning its proximity to decision boundaries. Analyzing localized neural collapse statistics could reveal whether spatial representations near the classification margin resist collapse significantly longer than those embedded deep within a specific class region. One more interesting direction

would be to look at how this works in the case of regression, as neural collapse has also been shown to happen in those tasks.
Furthermore, extending this framework to multi-class classification scenarios would provide deeper insights into how the Simplex Equiangular Tight Frame (ETF) manifests across complex, continuous spatial partitions. Additionally, while the flat torus embedding successfully isolated boundary artifacts in two dimensions, investigating whether these topological distortions scale to high-dimensional data manifolds remains a critical open question. Finally, exploring the impact of different architectural choices, such as convolutional networks, which inherently possess translation invariance, or alternative activation functions like periodic SIRENs could reveal how specific inductive biases alter the spatial eigenfunctions and the structural onset of Neural Collapse.

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